

Sniffing Out the Breed: Seeing What a Computer’s Nose Knows – Dog Breed Diffusion and Development

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Introduction:

The goal of the project is to see if a simple diffusion model can predict how mixed-breed dogs appear based on the combined signal (noise) of both trained breeds.

Input-output:

The model’s input is a simple text prompt denoting which breed(s) to generate an image of; the output is a generated image/batch of images.

Network and Design Details:

A UNet model was trained using standard diffusion training techniques. The input images were sized to 64x64. Unconditional training was performed first for 200 epochs; then text embeddings were added to the architecture for 80 additional epochs.

Dataset information:

The Stanford Dogs dataset [1], consisting of 20,580 images of 120 different breeds of dogs, was used for training the model; an additional, much smaller dataset from Kaggle, consisting of 339 mixed breed images and spanning 189 different breeds, was used only for visual analysis by us after the model was trained.

Evaluation:

CLIP scores [3] ranging between -1 and 1 expressed test image-prompt compatibility Evaluation was primarily qualitative, based on the quality of the images generated.

References:

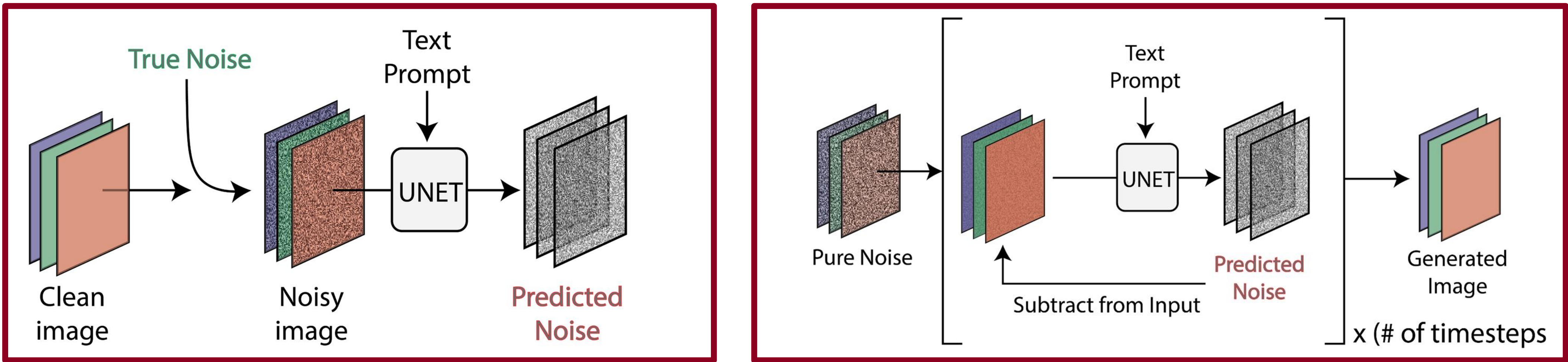
[1] Stanford Dog Dataset.
<https://www.kaggle.com/datasets/jessicali9530/stanford-dogs-dataset>

[2] Kaggle Mixed Dog Breed Dataset.
<https://www.kaggle.com/datasets/anjalikapoor14/mixed-breed-dog-dataset/data>

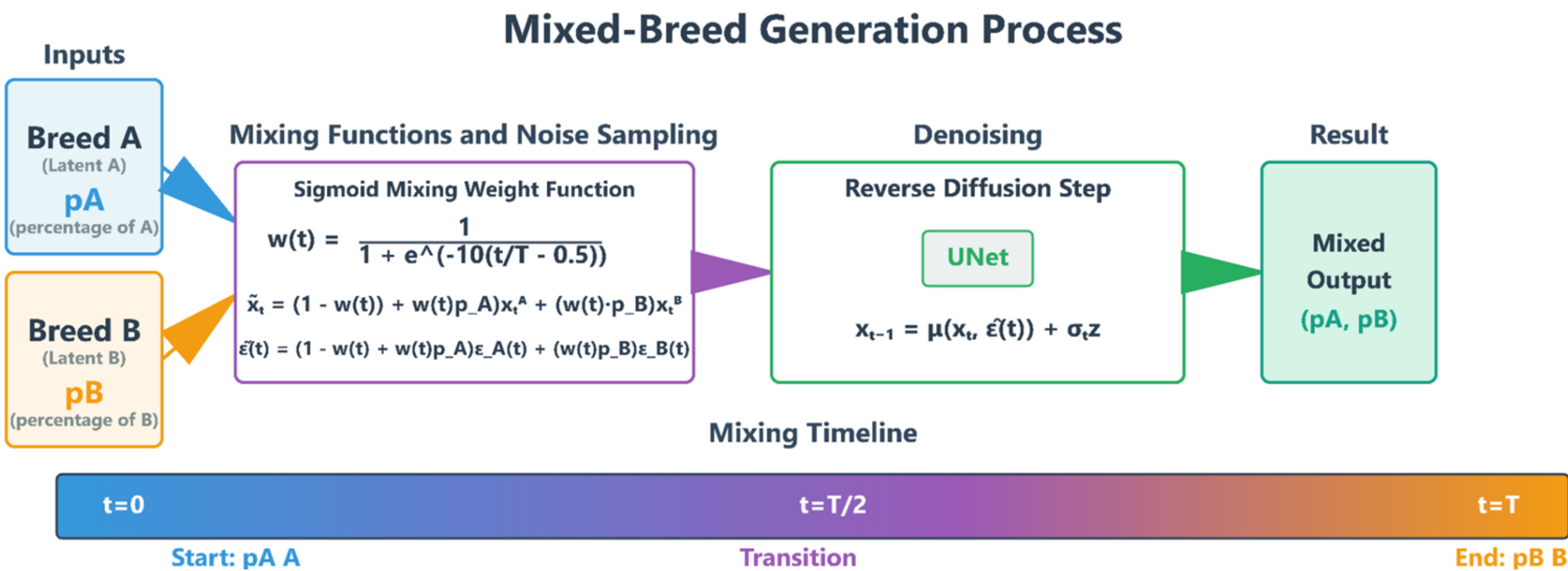
[3] Using CLIP Score to Evaluate Images.
<https://unimatrixz.com/blog/latent-space-clip-score/#clip-benchmarking>

Diffusion design is tough, but the results speak (or bark) for themselves

Diffusion Design

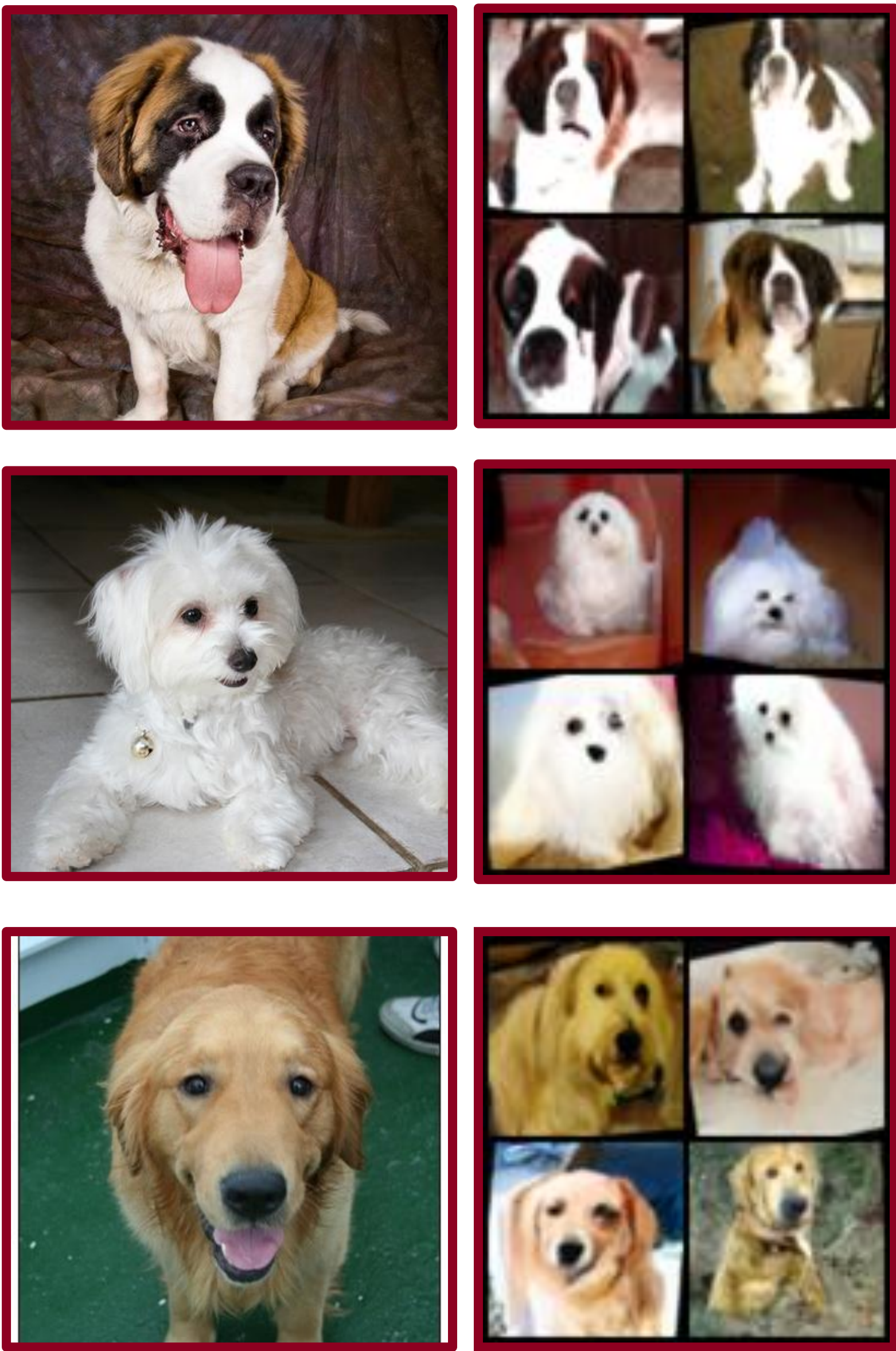


Mixed Breed Generation Methodology:



Model Results

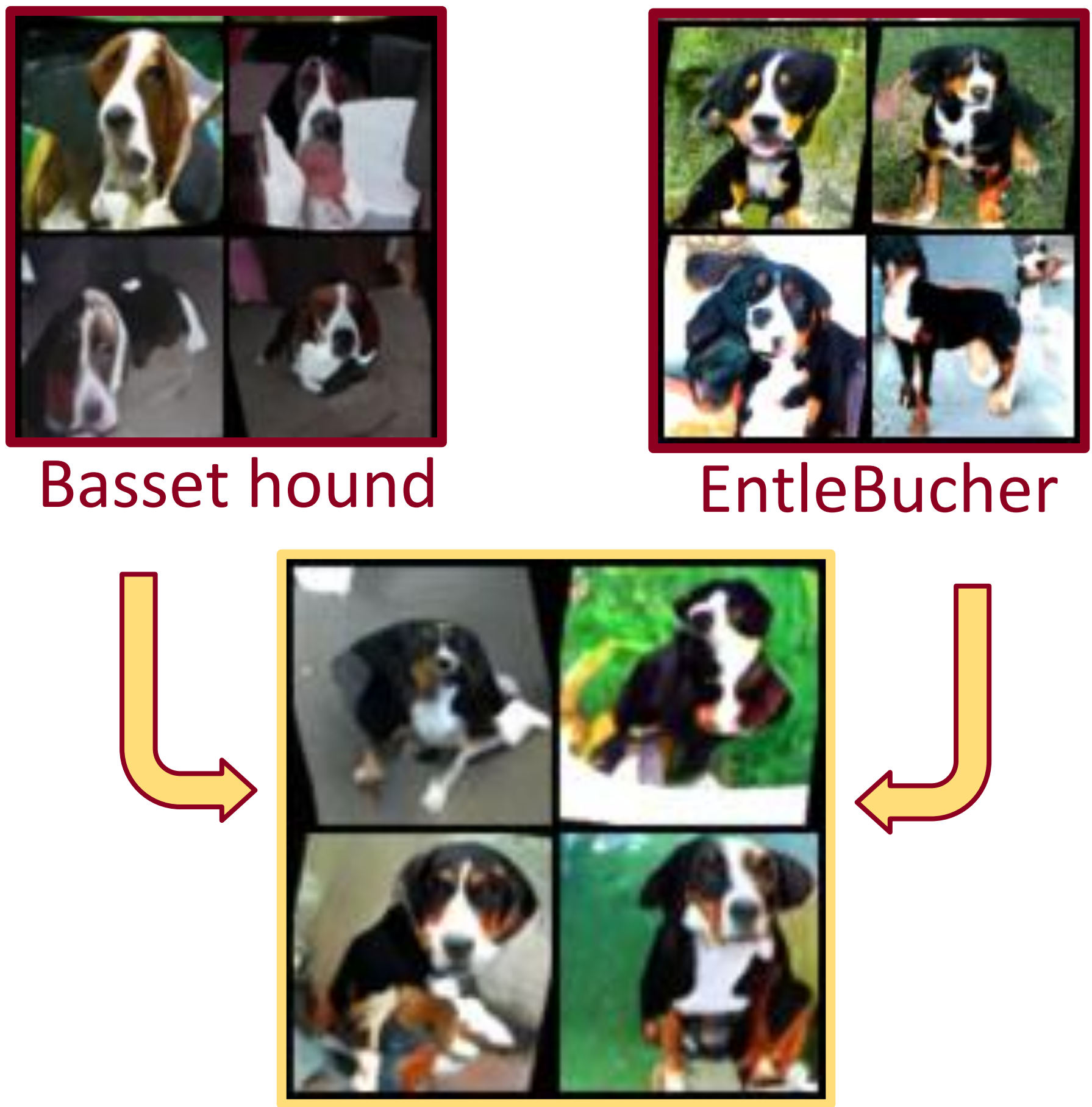
Single Breed



True
Images

Generated
Images

Mixed Breed



Quantitative Results:



Breed: Basset hound.
CLIP Score: 0.350

Breed: EntleBucher
CLIP Score: 0.362